

How the calculator prices a creator deal

This is the working explanation of the influencer calculator — the model behind both the standalone tool ([eyalcohen2524/influencer-calculator](#), pricing-engine upgrade in PR #1, engine in `public/pricing.js`, full math in `docs/PRICING_SPEC.md`) and the negotiation agent in this repo (`oblsk_negotiator/pricing.py`, the Python port that prices every move the agent makes). Same model, two frontends: the calculator is the interactive what-if tool, the agent runs the ladder live in a thread.

The one-sentence version: **estimate what a creator's posts will honestly deliver, translate delivery into dollars through our funnel, and derive three prices — where to open, where to aim, and the ceiling we never cross.**

1. The delivery estimate (and iterations on the old model)

The old model took the **median** of a creator's recent view counts and called that the expected delivery. Two failure modes:

- **It ignored the tail.** Creator content is heavy-tailed: most posts land near the usual mark, a few run 10–30x past it. A median throws that upside away entirely — and the occasional viral hit is a real part of why a multi-video deal is worth more than one video.
- **The old Monte Carlo had the opposite bug too:** it sampled *without* replacement up to the pool size and reported a median-of-sums, so the per-video estimate *rose* as you booked more videos (23,400 → 42,410 views per video as the count went 1 → 10 — an artifact, not a fact). It was also unseeded, so the headline number jittered on every unrelated slider move.

The new delivery model, per platform:

1. **Recency weighting** — each post is weighted by $0.5^{(\text{age}/90 \text{ days})}$, so a rising creator is not dragged down by year-old posts (and a fading one is not propped up by them).
2. **Winsorization (whale-capping)** — views above the 90th percentile are capped *for the planning number*, so one viral outlier cannot inflate the "typical campaign" estimate. In the spec's worked example this is the whole story: raw mean 71,623 views vs. an honest central estimate of 20,699.
3. **Viral tail, reported separately** — posts above 3x the median are surfaced as `viralProb` and `viralMean` (e.g. "6.7% of posts run to ~725k"). The hits aren't deleted; they're *upside*, not baseline. You plan on the central number and you tell the client the lottery ticket comes free.
4. **Organic → sponsored haircut (0.8)** — sponsored posts under-deliver a creator's organic history; the central estimate is scaled accordingly.
5. **Seeded bootstrap band** — future posts are independent draws *with* replacement from the capped pool, giving a reproducible p10–p90 band per campaign (same inputs, same numbers, always — a hard requirement for anything used in a negotiation).

6. **Cross-post dedup** — when the same video runs on TikTok and Instagram, summing platform views double-counts the overlapping audience; a 25% overlap haircut against the smaller platform yields unique reach.

The negotiation agent's port keeps the same ideas but runs them on its Monte Carlo: it fits a log-normal body with an explicit Pareto tail (so the upside is *modeled*, not just reported), applies the recency weights and the sponsored haircut in the fit, and winsorizes the simulated revenue draws when computing the expected value the ladder prices from. Downside = p10 of the simulation, upside = p90.

2. From views to dollars: the funnel

$\text{Value}(\text{views}) = \text{views} \times (\text{view} \rightarrow \text{signup rate}) \times (\text{signup} \rightarrow \text{paying rate}) \times \text{LTV} \times \text{margin}$

Applied to the downside / expected / upside delivery, this gives Value_down / Value_exp / Value_up — the deal's value under pessimistic, honest, and optimistic delivery. Every price below is derived from these three numbers, which is what makes each price *defensible*: change the funnel assumptions and every price moves with them, visibly.

3. The pricing ladder — the numbers you can defend to a client

Three independent ceilings:

Ceiling	Formula	Meaning
feeFromTargetRoi	$\text{Value_exp} / \text{ROI_target}$	the fee that hits our ROI goal (default 3x) at expected delivery
feeFromDownside	$\text{Value_down} / \text{ROI_min}$	the fee at which even the p10 downside still clears the minimum ROI (default 1x = break even in the bad case)
feeFromCpmCap	$\text{cap} \times \text{views_exp} / 1000$	an optional flat policy cap on \$ per 1,000 expected views

And from them, the ladder:

- **Anchor (where we open)** — ~62% of target, floored near market-low so the opening is credible rather than insulting. Justification to a creator: "this is what the expected delivery supports at our economics, with room to talk."
- **Target (fair fee at our goal)** — $\min(\text{feeFromTargetRoi}, \text{walkAway})$. The most we should *agree* to: at this fee, expected delivery returns our target multiple. This is the number to defend hardest.
- **Fair-market fee (context, not a commitment)** — the blended CPM benchmark $\text{mid} \times \text{expected views}$. Tells you where the market would put the deal. When market sits above target (common), the read is: "the market rate is payable, but our ROI-optimal number is lower — negotiate below market." When market sits below our anchor, our open is generous

and should close fast.

- **Break-even** — `Value_exp` itself: the fee at which expected ROI = 1.0. Useful in a client conversation as the absolute sanity line: *any* fee below this is expected-value positive; the question is only how positive.
- **Walk-away (never-exceed)** — `min(feeFromDownside, feeFromCpmCap)`. Above this, either a merely-unlucky campaign loses money or company CPM policy is broken. The engine reports which constraint is **binding**, so the negotiator knows which lever (ROI floor vs CPM policy) a stretch would actually bend.

Guaranteed by construction and by unit test: `anchor ≤ target ≤ walk-away`; ROI at target $\geq$ the goal whenever the ROI constraint binds; downside ROI at walk-away $\geq$ the floor whenever the downside binds.

The client script, in one line per number: "We open at \$X because that's what your expected reach supports at our economics; we're happy at \$Y, where the campaign pays 3-to-1; the market says \$M, which we can talk about; and we can't go past \$Z, because past that an ordinary underperformance — not a disaster, just a p10 post — puts the campaign underwater."

Worked example from the spec (a micro creator, 5 videos, real engine output): anchor **\$996** → target **\$1,366** (ROI 2.6x/3.0x/3.4x across down/expected/up) → fair market **\$2,104** → walk-away **\$3,609** (downside-ROI binding) → break-even **\$4,097**.

4. The ROI surface — showing risk, not a point estimate

For any fee, the engine reports ROI under downside, expected, and upside delivery. The negotiator (human or agent) sees a 3×3 grid — anchor / target / walk-away × down / expected / up — instead of one number. That's the direct answer to "what if the video flops?": at target, even the p10 outcome returns 2.6x in the worked example.

5. The authenticity heuristic (low-organic-engagement creators)

A brand pays for real, engaged reach; bought followers and engagement pods inflate the topline. Three cheap, transparent checks — no ML, every flag human-readable:

1. **Engagement per follower vs. tier norm** — (likes + comments) / followers against a floor that loosens with account size (1.5% under 100k followers, 1.0% to 1M, 0.6% above). Far below floor: -0.30; merely below: -0.12.
2. **Comments-to-likes ratio** — near zero (< 0.001) suggests like inflation or bots: -0.20.
3. **Views-to-followers** — median views under 3% of followers suggests a stale or purchased following: -0.15.

The score (clamped to [0.2, 1.0]) becomes a **risk discount on the downside**: `Value_down × score`. Because the walk-away ceiling is built from the downside, a weak-authenticity creator automatically gets a *tighter* ceiling — tested guarantee: lower authenticity $\Rightarrow$ strictly lower walk-away. The flags double as negotiation leverage ("engagement is 0.4% against a 1.5% norm for this tier — we have to price for that risk").

6. What to calibrate before trusting absolute dollars

Placeholders, in priority order (spec §13 and §15 have the full parameter table and data sources):

1. **Funnel numbers** (view → signup, signup → paying, LTV) — from attribution and product analytics. Everything scales linearly with these.
2. **CPM benchmark table** — currently rough US-market figures by platform and follower tier; replace with Oblsk's observed closed-deal rates.
3. **Sponsored factor (0.8)** — measure the organic → sponsored delivery ratio from past campaigns.
4. **Engagement floors** in the authenticity check — per-niche benchmarks (family/parenting content engages differently than gaming).

Relative comparisons (creator A vs. B, 1 video vs. 3) are robust before calibration; absolute dollar outputs are only as good as the funnel inputs.

7. Where the agent uses each piece

Calculator concept	In the negotiation agent
anchor	the opening flat offer
target	the one revised offer after pushback; the accept threshold early in a thread
walk-away	the accept ceiling late in a thread; asks above a multiple of it escalate to a person
binding constraint	reported in the ladder summary (<code>pricing.PriceLadder.summary()</code>)
authenticity → risk discount	<code>PricingPolicy.risk_discount</code> per creator
CPM cap	<code>cpm_cap_usd</code> in the campaign YAML
negotiation levers	the bundle escalation (volume is the buyer's strongest lever) and the fast-pay sweetener

Backlog note: the calculator's usage-rights premium is not yet a first-class input in either codebase — currently usage terms live in the campaign brief text. A usage-rights toggle on the calculator (and a corresponding fee bump) is a planned addition.